

02LQD Specification and Simulation of Digital Systems

2018/02/23

1. Combinational circuit design (5 points)

Excess-3 is a self-complementary, biased and redundant code. Each decimal digit is expressed on 4 bits. The following table matches BCD and excess-3:

BCD	excess-3
0000	0011
0001	0100
0010	0101
0011	0110
0100	0111
0101	1000
0110	1001
0111	1010
1000	1011
1001	1100

Design in VHDL a BCD to excess-3 transcoder implementing the above table according to the following description styles:

1. dataflow using conditional signal assignments (WHEN ELSE statements)
2. dataflow using selected signal assignments (WITH SELECT WHEN statements)

2. FSM-D design (25 points)

Fibonacci numbers are recursively defined as follows:

$$FIB_n = FIB_{n-2} + FIB_{n-1} \quad n > 1$$

$$FIB_0 = 0$$

$$FIB_1 = 1$$

Design a FSM-D to **iteratively** compute Fibonacci numbers of order n . Note that n is on 4 bits. Note that, being $FIB_{15}=610$, the output `data_out` is on 10 bits. Computation starts when a `start` signal is asserted for one clock cycle. When computation is over the `ready` signal is asserted for one clock cycle. Reset is asynchronous. The output `data_out` makes sense only when `ready` is 1.

Steps:

1. Draw the HLSM state diagram and write the VHDL code to capture the HLSM behavior (10 points)
2. Design a FSM-D (15 points):
 - create 2 processes for behavioural datapath description from the HLSM state diagram
 - create 2 processes for behavioural controller description from the HLSM state diagram
 - create a process for behavioral output description from the HLSM state diagram.

Exam rules:

- upload on Portale della didattica in section Elaborati:
 - running VHDL code and testbench
 - short report (max 3 pages) describing the solution highlighting differences w.r.t. the solution handed in at the written exam
- no later than Monday February 26 12.00 am**
- remember: no upload $\Rightarrow$ no correction $\Rightarrow$ no oral exam
 - once the ranking is published on Portale della didattica in section Materiale/Results, drop an e-mail to paolo.camurati@polito.it to agree on a date for the oral exam.